

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Ethical Hacking

COURSE CODE: ITA14

COURSE OUTCOME COVERED

CO2: Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT4)

Assessment Tool Weightage: 30%

Duration: 90 min

Total Marks: 50

Social Engineering Simulation

Questions

1. As a Security Analyst, perform an OSINT-based email enumeration assessment on the target domain **tesla.com** using **theHarvester** in Kali Linux. Collect publicly available email addresses, subdomains, hosts, and IP addresses associated with the organization. Categorize the discovered email addresses into likely roles such as Support, Sales, Human Resources, Administration, and Technical Teams. Analyze how the gathered information could assist an attacker during the reconnaissance phase and explain how it could be used for spear-phishing, credential harvesting, and impersonation attacks. Finally, recommend at least five countermeasures to reduce information leakage and social engineering risks, and submit a report with screenshots, commands executed, findings, risk analysis, and recommendations.

Sol : To perform an OSINT-based reconnaissance on **tesla.com** using theHarvester in Kali Linux and identify publicly available information such as subdomains, hosts, email addresses, and IP addresses.

Command 1 : theHarvester -d tesla.com -b rapiddns

Purpose :

This command performs an OSINT scan on the target domain tesla.com using the RapidDNS search engine. It collects publicly available information such as email addresses, subdomains, hosts, and IP addresses.

```
Session Actions Edit View Help
zsh: corrupt history file /home/kali/.zsh_history
kali@kali: ~$ theHarvester -d tesla.com -b rapiddns
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
theHarvester
*****
* theHarvester 4.8.2
* Coded by Christian Martorella
* Edge-Security Research
* chr00r@edge-security.com
*****

[*] Target: tesla.com
[*] Searching RapidDns.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] Hosts found: 88

akamai-apigateway-automation-billing.tesla.com:akamai-apigateway-automation-billing.tesla.com.edgekey.net
akamai-apigateway-automation.tesla.com:akamai-apigateway-automation.tesla.com.edgekey.net
akamai-apigateway-captiveunderwriting.tesla.com:akamai-apigateway-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-dev-warptmsapiserver.tesla.com:akamai-apigateway-dev-warptmsapiserver.tesla.com.edgekey.net
akamai-apigateway-logisticratesapi.tesla.com:akamai-apigateway-logisticratesapi.tesla.com.edgekey.net
akamai-apigateway-precuretopayapi.tesla.com:akamai-apigateway-precuretopayapi.tesla.com.edgekey.net
akamai-apigateway-shipmentplanningapi.tesla.com:akamai-apigateway-shipmentplanningapi.tesla.com.edgekey.net
akamai-apigateway-stg-captiveunderwriting.tesla.com:akamai-apigateway-stg-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-stg-deliveryopsapi.tesla.com:akamai-apigateway-stg-deliveryopsapi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-defi.tesla.com:akamai-apigateway-stg-finplateng-defi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-routene.tesla.com:akamai-apigateway-stg-finplateng-routene.tesla.com.edgekey.net
akamai-apigateway-stg-fra.tesla.com:akamai-apigateway-stg-fra.tesla.com.edgekey.net
akamai-apigateway-stg-inventorynextapi.tesla.com:akamai-apigateway-stg-inventorynextapi.tesla.com.edgekey.net
akamai-apigateway-stg-translationpipeline.tesla.com:akamai-apigateway-stg-translationpipeline.tesla.com.edgekey.net
akamai-apigateway-vfx.tesla.com:akamai-apigateway-vfx.tesla.com.edgekey.net
akamai-apigateway-warptmsapi.tesla.com:akamai-apigateway-warptmsapi.tesla.com.edgekey.net
auth.tesla.com:auth.tesla.com.edgekey.net
autobidder-preprod.powerhub.energy.tesla.com:energy.tesla.com.edgekey.net
autodiscover.tesla.com:autodiscover.outlook.com
bi.tesla.com:ipa.teslartz.net.srip.net
billing.tesla.com:billing.tesla.com.edgekey.net
```

```
Session Actions Edit View Help
business-ui-ownership.tesla.com:business-ui-ownership.tesla.com.edgekey.net
cloudprotect.tesla.com:tesla.gncloudservice.com
civn.tesla.com:10.222.41.215
dataviz.tesla.com:ipa.teslartz.net.srip.net
edr.tesla.com:edr.tesla.com-vl.edgekey.net
elivoincing.tesla.com:external-na-pop1.edgekey.net
employeefeedback.tesla.com:employeefeedback.tesla.com.akadns.net
engage.tesla.com:2617ba835e2.cloudfront.net
envoy-partnerleadsharing.tesla.com:external-na-pop1.edgekey.net
epuca-prd.usw.vn.cloud.tesla.com:epuca-prd-a-defa38e7209f44a.elb.eu-west-1.amazonaws.com
epuca-prd.usw.vn.cloud.tesla.com:usw-prd5a-fleet-epuca-prd.teslafleet.akadns.net
epuca-prd.usw.vn.cloud.tesla.com:usw-prd5b-fleet-epuca-prd.teslafleet.akadns.net
epuca-prd.usw.vn.cloud.tesla.com:usw-prd5c-fleet-epuca-prd.teslafleet.akadns.net
epuca-prd.usw.vn.cloud.tesla.com:usw-prd5d-fleet-epuca-prd.teslafleet.akadns.net
epuca-prd.usw2.vn.cloud.tesla.com:epuca-prd-a-a391d835e78d04f.elb.us-west-2.amazonaws.com
epuca-prd.usw2.vn.cloud.tesla.com:epuca-prd-b-2e18d6967d0d51.elb.us-west-2.amazonaws.com
epuca-prd.usw2.vn.cloud.tesla.com:epuca-prd-c-d59ac96c73767a53.elb.us-west-2.amazonaws.com
erilog.tesla.com:erilog.tesla.com.edgekey.net
forums.tesla.com:site-6030136.onvanilla.net
grid.tesla.com:grid.tesla.com.edgekey.net
hermes-app-eng.vn.cloud.tesla.com:usw-eng5x-hermes-npuaph-prd.teslafleet.akadns.net
image-emails.tesla.com:image-emails.tesla.com.edgekey.net
image-email.tesla.com:image-sf.exacttarget.com.edgesuite.net
links.tesla.com:links.tesla.com.edgekey.net
livestreamapi.tesla.com:livestreamapi.tesla.com.edgekey.net
rfs-supplier.tesla.com:rfs-supplier.tesla.com.edgekey.net
mobile.tesla.com:mobile.tesla.com.edgekey.net
ota_email.tesla.com:13.111.390
npuhe-eng.euw1.vn.cloud.tesla.com:npuhe-eng-c-28d6e95ea80c5e2a.elb.eu-west-1.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-c-63f72d0c31dca.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-b-979131c1931a27b.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-c-ef47d5b3ed829c9b.elb.us-west-2.amazonaws.com
npuhe-eng.usw.vn.cloud.tesla.com:usw-eng5a-fleet-npuhe-prd.teslafleet.akadns.net
npuhe-eng.usw.vn.cloud.tesla.com:usw-eng5c-fleet-npuhe-prd.teslafleet.akadns.net
npuhe-prd.usw.vn.cloud.tesla.com:usw-prd5b-fleet-npuhe-prd.teslafleet.akadns.net
npuhe-prd.usw2.vn.cloud.tesla.com:npuhe-prd-a-e02568703bea82a.elb.us-west-1.amazonaws.com
npuv-ap-prd.euw1.vn.cloud.tesla.com:npuv-prd-euw1.vn.cloud.tesla.com
npuv-eng.euw1.vn.cloud.tesla.com:npuv-eng-a-38aef4915757d5fc.elb.eu-west-1.amazonaws.com
npuv-eng.usw.vn.cloud.tesla.com:usw-eng5d-fleet-npuv-prd.teslafleet.akadns.net
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-a-97a83e9f6d010b4.elb.us-west-2.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-a-56b258e6f051aaf.elb.us-west-1.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-c-04e07fce0e0d2bb.elb.us-west-1.amazonaws.com
npuv-prd.usw.vn.cloud.tesla.com:npuv-prd-fleet-npuv-prd.teslafleet.akadns.net
npuv-prd.usw2.vn.cloud.tesla.com:npuv-prd-c-a2aa8526360f4cf.elb.us-west-2.amazonaws.com
origin-bolt.tesla.com:clandecpp.tesla.com.akadns.net
origin-courses.tesla.com:origin-courses.tesla.com.akadns.net
origin-edr.tesla.com:cligenp.tesla.com.akadns.net
origin-elivoincing.tesla.com:warpp-elivoincing.tesla.com.akadns.net
origin-finops.tesla.com:clandcfiap.tesla.com.akadns.net
origin-livestream.tesla.com:clcligen.tesla.com.akadns.net
origin-location-services-prd.tesla.com:clandcvhdp.tesla.com.akadns.net
```

Command 2 : theHarvester -d tesla.com -b rapiddns -n

Purpose :
This command performs DNS lookup on the discovered hosts to retrieve DNS-related information and verify the discovered domains.

```
Session Actions Edit View Help
links.tesla.com:links.tesla.com.edgekey.net
livestreamapi.tesla.com:livestreamapi.tesla.com.edgekey.net
mfs-supplier.tesla.com:mfs-supplier.tesla.com.edgekey.net
mobile.tesla.com:mobile.tesla.com.edgekey.net
mta.email.tesla.com:13.111.14.198
npuhe-eng.euw1.vn.cloud.tesla.com:npuhe-eng-c-286de95ea00c5e2a.elb.eu-west-1.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-b-979131c19311427b.elb.us-west-2.amazonaws.com
npuhe-eng.usw2.vn.cloud.tesla.com:npuhe-eng-c-ef07d5b3ed829c9b.elb.us-west-2.amazonaws.com
npute-eng.usw.vn.cloud.tesla.com:usw-eng5a-fleet-npute-prd.teslafleet.akadns.net
npute-eng.usw.vn.cloud.tesla.com:usw-eng5c-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw.vn.cloud.tesla.com:usw-prd5b-fleet-npute-prd.teslafleet.akadns.net
npute-prd.usw2.vn.cloud.tesla.com:npute-prd-a-ea2568703deac02e.elb.us-west-2.amazonaws.com
npuv-ap-prd.euw1.vn.cloud.tesla.com:npuv-prd.euw1.vn.cloud.tesla.com
npuv-eng.euw1.vn.cloud.tesla.com:npuv-eng-a-38adfa93f5795d7c.elb.eu-west-1.amazonaws.com
npuv-eng.usw2.vn.cloud.tesla.com:npuv-eng-a-97483e916dd01bb4.elb.us-west-2.amazonaws.com
npuv-prd.euw1.vn.cloud.tesla.com:npuv-prd-a-56bc286e6fb51aaf.elb.eu-west-1.amazonaws.com
npuv-prd.usw.vn.cloud.tesla.com:usw-prd5c-fleet-npuv-prd.teslafleet.akadns.net
npuv-prd.usw2.vn.cloud.tesla.com:npuv-prd-c-a2aa8526c360f4cf.elb.us-west-2.amazonaws.com
origin-bolt.tesla.com:clsmdcgyp.tesla.com.akadns.net
origin-courses.tesla.com:origin-courses.tesla.com.akadns.net
origin-edr.tesla.com:clsgenp.tesla.com.akadns.net
origin-invoicing.tesla.com:warpp-invoicing.tesla.com.akadns.net
origin-finops.tesla.com:clsmdcfinp.tesla.com.akadns.net
origin-livestream.tesla.com:klsgenp.tesla.com.akadns.net
origin-location-services-prd.tesla.com:clsmdcvehp.tesla.com.akadns.net
origin-static-assets-profile-settings.tesla.com:clsmdcgenp.tesla.com.akadns.net
origin-tg-gw.stg.tesla.com:edgeservices.tesla.com.akadns.net
origin-warpp.akamai.tesla.com:199.66.9.77
partners.tesla.com:partners.tesla.com.edgekey.net
sales-pa.tesla.com:xtg-ext-na-pop1.edgekey.net
shop.tesla.com:shop.tesla.com.edgekey.net
static-assets-pay.tesla.com:static-assets-pay.tesla.com-v1.edgekey.net
static-edr.tesla.com:static-assets.tesla.com.edgekey.net
stg-uc-twilio-bpm-webhook.tesla.com:external-na-pop1.edgekey.net
suppliers.tesla.com:suppliers.tesla.com.edgekey.net
toolbox.tesla.com:toolbox.tesla.com.edgekey.net
trt.tesla.com:external-na-pop1.edgekey.net
ul.engage.tesla.com:d1e601a26qsud.cloudfront.net
url2547.tesla.com:sendgrid.net
url15196.tesla.com:sendgrid.net
url17051.tesla.com:url17051.tesla.com.edgekey.net

[*] Starting active queries for DNSLookup.
[*] Hosts found after reverse lookup (in target domain):
```

```
Session Actions Edit View Help
url17051.tesla.com:url17051.tesla.com.edgekey.net

(kali@kali) ~ - ssh
$ theHarvester -d tesla.com -b rapiddns -n
Read proxies.yaml from /etc/theHarvester/proxies.yaml
*****
*  TheHarvester
*
*  theHarvester 4.18.1
*  Coded By Christian Martorella
*  Edge-Security Research
*  cmartorella@edge-security.com
*
*****

[*] Target: tesla.com
[*] Searching RapidDns.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] Hosts found: 88
akamai-apigateway-automation-billing.tesla.com:akamai-apigateway-automation-billing.tesla.com.edgekey.net
akamai-apigateway-automation.tesla.com:akamai-apigateway-automation.tesla.com.edgekey.net
akamai-apigateway-dev-warptesapiserver.tesla.com:akamai-apigateway-dev-warptesapiserver.tesla.com.edgekey.net
akamai-apigateway-logisticsratesapi.tesla.com:akamai-apigateway-logisticsratesapi.tesla.com.edgekey.net
akamai-apigateway-procuretopayapi.tesla.com:akamai-apigateway-procuretopayapi.tesla.com.edgekey.net
akamai-apigateway-shipmentplanningapi.tesla.com:akamai-apigateway-shipmentplanningapi.tesla.com.edgekey.net
akamai-apigateway-stg-captiveunderwriting.tesla.com:akamai-apigateway-stg-captiveunderwriting.tesla.com.edgekey.net
akamai-apigateway-stg-deliveryopsapi.tesla.com:akamai-apigateway-stg-deliveryopsapi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-defi.tesla.com:akamai-apigateway-stg-finplateng-defi.tesla.com.edgekey.net
akamai-apigateway-stg-finplateng-routeone.tesla.com:akamai-apigateway-stg-finplateng-routeone.tesla.com.edgekey.net
akamai-apigateway-stg-fits.tesla.com:akamai-apigateway-stg-fits.tesla.com.edgekey.net
akamai-apigateway-stg-inventorynextapi.tesla.com:akamai-apigateway-stg-inventorynextapi.tesla.com.edgekey.net
akamai-apigateway-stg-translationpipeline.tesla.com:akamai-apigateway-stg-translationpipeline.tesla.com.edgekey.net
akamai-apigateway-vfx.tesla.com:akamai-apigateway-vfx.tesla.com.edgekey.net
akamai-apigateway-warptapi.tesla.com:akamai-apigateway-warptapi.tesla.com.edgekey.net
auth.tesla.com:auth.tesla.com.edgekey.net
autobidder-preprd.powerhub.energy.tesla.com:energy.tesla.com.edgekey.net
autodiscover.tesla.com:autodiscover.outlook.com
```

Findings

Target Domain: tesla.com

Information Collected

Item	Result
Email Addresses	No public email addresses found
IP Addresses	No public IP addresses found
Hosts/Subdomains	88 Hosts Found
People	No people found

Risk Analysis

The discovered subdomains provide useful reconnaissance information about Tesla's public infrastructure. An attacker could use these hostnames to understand exposed services, identify potential targets, and prepare phishing or impersonation attacks. Although no public email addresses were found, the discovered hosts can still support reconnaissance when combined with other OSINT sources.

Countermeasures

1. Conduct regular OSINT assessments to identify exposed information.
2. Enable Multi-Factor Authentication (MFA).
3. Train employees to recognize phishing attacks.
4. Reduce unnecessary exposure of public subdomains.
5. Continuously monitor public-facing assets and DNS records.

Conclusion :

The OSINT assessment successfully identified **88 public hosts/subdomains** related to tesla.com. No public email addresses or IP addresses were discovered. The collected information demonstrates how attackers can gather infrastructure details during the reconnaissance phase and highlights the importance of minimizing public information exposure and implementing strong security controls.

2. As a cybersecurity awareness consultant, you are required to conduct a phishing awareness simulation using GoPhish and dummy employee accounts provided in the lab. Create an email template, launch a campaign, monitor user interactions, and generate a report showing delivery rate, email open rate, and click rate. Based on the results, provide recommendations to improve user awareness against phishing attacks.

Command 1 : sudo apt update

Purpose :

Updates the Kali Linux package repository to ensure the latest package information is available before installing software.

```
kali@kali: ~/gophish
Session Actions Edit View Help
Error: Some index files failed to download. They have been ignored, or old ones used instead.

(kali@kali) [-]
$ sudo apt clean

(kali@kali) [-]
$ sudo rm -rf /var/lib/apt/lists/*

(kali@kali) [-]
$ sudo apt update
get1 http://mirrors.esto.network/kali kali-rolling InRelease [36.0 kB]
get12 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [21.4 MB]
get13 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents [53.2 MB]
get14 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Packages [185 kB]
get15 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Contents [deb] [189 kB]
get16 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Packages [69 kB]
get17 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Contents [deb] [888 kB]
get18 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Packages [15.8 kB]
get19 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Contents [deb] [46.5 kB]
Fetched 76.0 MB in 24s (3,153 kB/s)
670 packages can be upgraded. Run 'apt list --upgradable' to see them.

(kali@kali) [-]
$ sudo apt install wget unzip -y
[sudo] password for kali:
unzip is already the newest version (6.0-29).
unzip set to manually installed.
Upgrading:
  wget
Summary:
Upgrading: 1, Installing: 0, Removing: 0, Not Upgrading: 869
Download size: 985 kB
Space needed: 0 B / 62.8 GB available

get1 http://mirrors.esto.network/kali kali-rolling/main amd64 wget amd64 1.25.0-3 [985 kB]
Fetched 985 kB in 20s (49.2 kB/s)
Reading database... 629/76 files and directories currently installed.)
Preparing to unpack ./wget-1.25.0-3_amd64.deb...
Unpacking wget (1.25.0-3) over (1.25.0-2)...
Setting up wget (1.25.0-3)...
Processing triggers for kali-menu (2026.2.6)...
Processing triggers for man-db (2.11.1-2)...

(kali@kali) [-]
$ wget https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
--2026-08-05 03:14:54-- https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip
```

Command 2 : `wget https://github.com/gophish/gophish/releases/latest/download/gophish-v0.12.1-linux-64bit.zip`

Purpose :

Downloads the latest GoPhish Linux package from the official GitHub repository.

[illegible]

Command 3 : unzip gophish-v0.12.1-linux-64bit.zip -d gophish

Command 4 : cd gophish

Command 5 : chmod +x gophish

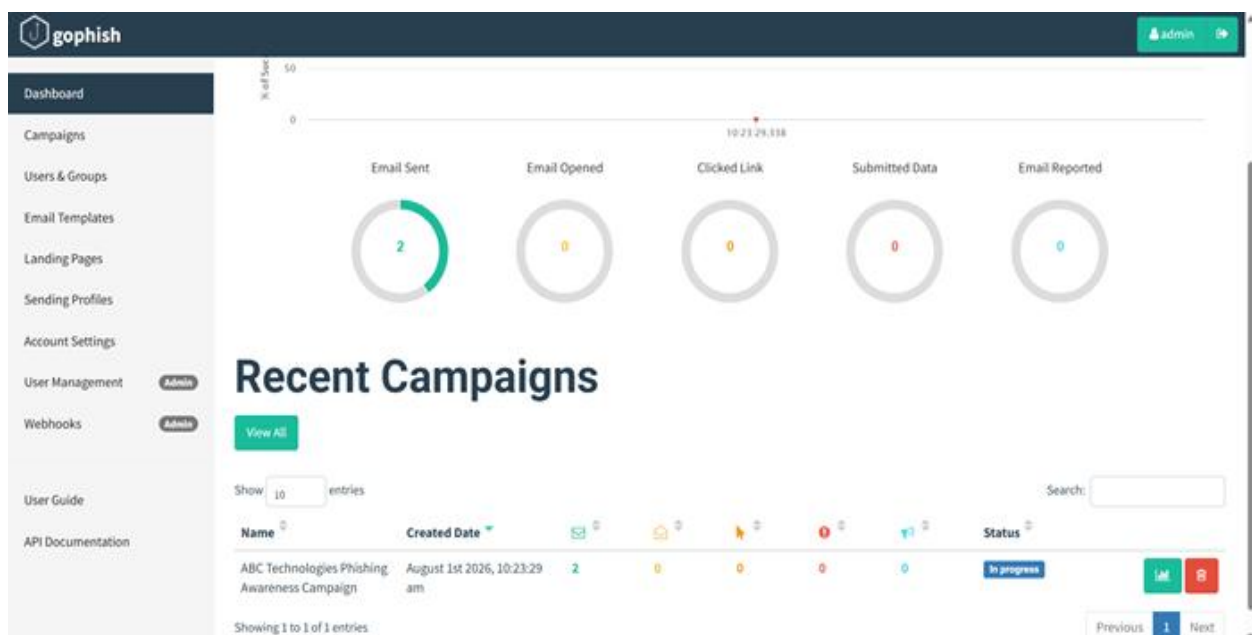
Command 5 : sudo ./gophish

Purpose :

Extracts the downloaded GoPhish ZIP archive into the **gophish** directory. Changes the current working directory to the GoPhish installation folder. Grants execute permission to the GoPhish executable so that it can be started. Starts the GoPhish application, initializes the database, generates SSL certificates if required, and launches the phishing server and the administration web interface.

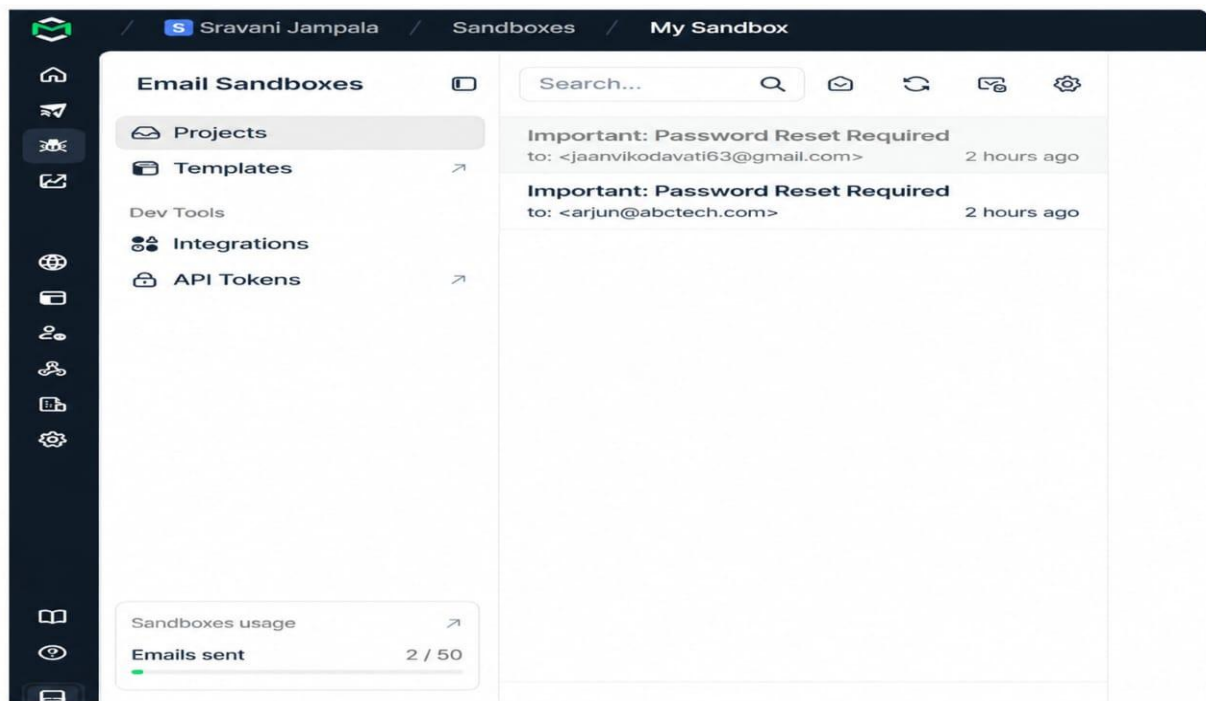


ABC - Technologies
Phishing Awareness.xl



The screenshot shows the GoPhish dashboard with the 'Details' view selected. The top navigation bar and left sidebar are the same as in the previous screenshot. The main content area displays five circular progress charts for campaign statistics: Email Sent (2), Email Opened (0), Clicked Link (0), Submitted Data (0), and Email Reported (0). Below these charts is a section titled 'Details' with a 'Show 10 entries' dropdown and a search bar. A table lists the details of the campaign, with columns for First Name, Last Name, Email, Position, Status, and Reported. The table has five entries, with the first entry being 'sravani jampala' with email 'sravani.jampala@abctech.com' and status 'Email Sent'. The bottom of the dashboard shows 'Showing 1 to 5 of 5 entries' and navigation buttons for 'Previous', '1', and 'Next'.

First Name	Last Name	Email	Position	Status	Reported
sravani	jampala	sravani.jampala@abctech.com		Email Sent	
vikram	reddy	vikram.reddy@abctech.com		Error	
ananya	desai	ananya.desai@abctech.com		Error	
rohit	mehra	rohit.mehra@abctech.com		Error	
kavya	nair	kavya.nair@abctech.com		Email Sent	



Email Security Controls :

- Enable Spam Filtering.
- Use Email Gateways.
- Configure SPF (Sender Policy Framework).
- Configure DKIM (DomainKeys Identified Mail).
- Configure DMARC (Domain-based Message Authentication, Reporting, and Conformance).
- Block dangerous attachments.
- Scan email attachments using antivirus software.
- Enable URL filtering.

3. Using **Nmap Aggressive Scan (-A)** on **scanme.nmap.org**, gather detailed information about the target including operating system details, open ports, and service versions. Assume you are an attacker and describe how the collected information could be leveraged to conduct a social engineering attack against the IT support team. Prepare a report explaining the attack methodology and defensive recommendations.

Sol : To perform an aggressive scan on **scanme.nmap.org** using Nmap to identify the target operating system, open ports, running services, and service versions. The collected information is analyzed to understand how an attacker could use it in a social engineering attack against the IT support team and to recommend defensive measures.

Command 1 : `nmap -A scanme.nmap.org`

Purpose

Performs an **Aggressive Scan** on the target. The -A option enables:

- Operating System Detection
- Service Version Detection
- Script Scanning (NSE)
- Traceroute

```
zsh: corrupt history file /home/kali/.zsh_history
kali@kali:~$ nmap -A scanme.nmap.org
Starting Nmap 7.95 ( https://nmap.org ) at 2020-08-05 12:57 EDT
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.802s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01:f03c:93ff:fe18:b62f
Not shown: 989 filtered tcp ports (no-response)
PORT      STATE SERVICE
22/tcp    open  tcpwrapped
139/tcp   closed smb
53/tcp    closed domain
80/tcp    open  http    Apache httpd 2.4.7 ((Ubuntu))
139/tcp   closed netbios-ssn
443/tcp   closed https
3389/tcp  closed ms-wbt-server
5986/tcp  open  sip?
5985/tcp  open  sip-tls?
8080/tcp  closed http-proxy
8081/tcp  closed blackice-icecap
Device type: switch
Running (JUST GUESSING): Cisco embedded (85%)
OS CPE: cpe:/o:linux:linux-gnu
Aggressive OS guesses: Cisco SG 500 switch (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.89 ms scanme.nmap.org (45.33.32.156)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/.
Nmap done: 1 IP address (1 host up) scanned in 93.84 seconds

kali@kali:~$
```

Findings :

Parameter	Result
Target Host	scanme.nmap.org
Target IP Address	45.33.32.156
Host Status	Up
Network Distance	1 Hop
Device Type	VoIP Adapter / Bridge / General Purpose

port	state	service	version
------	-------	---------	---------

22	Open	tcpwrapped	SSH Host
80	Open	HTTP	Apache HTTP Server
5060	Open	SIP	SIP Service
5061	Open	SIP-TLS	Secure SIP Service

Attack Methodology

Step 1 - Reconnaissance

The attacker performs an Nmap Aggressive Scan to gather information about the target system, including operating system details, open ports, and running services.

Step 2 - Service Enumeration

The scan reveals:

- Apache HTTP Server 2.4.7 running on Ubuntu
- SSH service
- SIP and SIP-TLS services

Step 3 - Social Engineering Attack

The attacker contacts the IT support team pretending to be a network administrator or software vendor.

Step 4 - Credential Theft

If the IT support team trusts the attacker, they may reveal usernames, passwords, VPN details, or other sensitive information.

Risks

- Exposure of server information
- Credential theft
- Unauthorized system access

Conclusion

The Nmap Aggressive Scan successfully identified the target's IP address, open ports, running services, and probable operating system details. Such information can be valuable during the reconnaissance phase of an attack and may be used to conduct convincing social engineering attacks against IT support personnel. Implementing strong security controls, regular patch management, and employee awareness training can significantly reduce these risks.

Code of Conduct:

I Jampala Sravani - 192372141 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jampala Sravani

RUBRICS FOR CALCULATION**Social Engineering Simulation**

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Social Engineering Scenario	20%	Demonstrates a comprehensive understanding of the social engineering scenario, attack techniques, and security context	Demonstrates good understanding with minor omissions.	Shows partial understanding but misses important aspects of the scenario.	Fails to understand or incorrectly interprets the social engineering scenario.
Identification of Attack and Vulnerabilities	20%	Correctly identifies the attack type, exploited vulnerabilities, and affected assets with complete accuracy	Correctly identifies most attack techniques with minor errors.	Identifies only some attack methods and vulnerabilities.	Unable to identify the attack type or associated vulnerabilities.
Application of Security Concepts and Countermeasures	25%	Effectively applies cybersecurity concepts and recommends appropriate preventive, detective, and corrective countermeasures.	Applies relevant concepts with minor errors and recommends suitable countermeasures.	Applies concepts partially with limited recommendations.	Unable to apply relevant cybersecurity concepts or recommend suitable countermeasures.
Risk Analysis and Decision Making	20%	Performs logical risk analysis, evaluates business impact, and provides well-justified security decisions.	Provides reasonable analysis with adequate justification.	Performs limited analysis with weak justification.	Provides no meaningful analysis or justification.
Clarity, Documentation and Technical Presentation	15%	Response is well-organized, technically accurate, professionally documented, and logically presented.	Response is clear with minor documentation or presentation issues.	Response lacks organization and contains technical inaccuracies.	Response is poorly organized, incomplete, and difficult to understand.